Multivariable calculus

1. Cross and dot product

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}, \quad \left|\vec{a} \times \vec{b}\right| = |\vec{a}|\left|\vec{b}\right| \sin\theta$$

$\vec{a} \times \vec{b}$ is orthogonal to both; direction by right-hand rule; magnitude = area of parallelogram spanned by $\vec{a}$ and $\vec{b}$.

2. Planes

Vector equation (normal $\vec{n}$, point $\vec{P}$): $\vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{P}$

Normal to level surface $F(\vec{x}) = k$ at $\vec{a}$: $\nabla F(\vec{a})$; tangent plane: $\nabla F(\vec{a}) \cdot \vec{r} = \nabla F(\vec{a}) \cdot \vec{a}$

Plane through three points A, B, C: normal $\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC}$

3. Differentiation

Gradient: $\nabla f(\vec{x}) = \langle \frac{\partial f}{\partial x_1}, \cdots, \frac{\partial f}{\partial x_n} \rangle$; perpendicular to level sets; points toward steepest ascent with magnitude $|\nabla f| = \max_{\hat{v}} D_{\hat{v}} f$.

Jacobian of $\vec{f} : \mathbb{R}^n \to \mathbb{R}^m$ at $\vec{x}$ (outputs as rows, inputs as columns):

$$\mathrm{J}\vec{f}(\vec{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

For scalar $f : \mathbb{R}^n \to \mathbb{R}$: $\mathrm{J}f = (\nabla f)^{\mathrm{T}}$ (a row vector).

Linear approximation to $f : \mathbb{R}^n \to \mathbb{R}$ at $\vec{a}$ (tangent hyperplane to graph):

$$L = f + \nabla f \cdot \Delta\vec{x}$$
$$L(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a})$$

Chain rule (generic): if $\vec{r} : \mathbb{R}^n \to \mathbb{R}^m$ and $\vec{f} : \mathbb{R}^m \to \mathbb{R}^p$:

$$\mathrm{J}\left(\vec{f} \circ \vec{r}\right) = \mathrm{J}\vec{f}\mathrm{J}\vec{r}$$

In $\mathbb{R}^3$, $f \circ \vec{r} : T \subseteq \mathbb{R} \to \mathbb{R}^3 \to \mathbb{R}$), with $(x, y, z) \coloneqq (r_1, r_2, r_3)$:

$$\frac{d(f \circ \vec{r})}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}.$$

Directional derivative: $D_{\hat{v}} f(\vec{a}) = \nabla f(\vec{a}) \cdot \hat{v}$ (rate of change in direction $\hat{v}$).

Clairaut: $f_{xy} = f_{yx}$ when both mixed partials are continuous.

4. Curves

Arc length: $L = \int_a^b \left|\frac{\mathrm{d}\vec{r}}{\mathrm{d}t}\right| \mathrm{d}t$; arc length function $s(t) = \int_a^t |\vec{r}'(u)| \,\mathrm{d}u$

Unit tangent: $\vec{T} = \hat{r'} = \vec{r}'/|\vec{r}'|$

Curvature (rate of direction change per unit arc length):

$$\kappa = \left|\frac{\mathrm{d}\vec{T}}{\mathrm{d}s}\right| = \frac{\left|\vec{T}'\right|}{|\vec{r}'|} = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$$

Principal unit normal: $\vec{N} = \widehat{T'} = \vec{T}'/\left|\vec{T}'\right|$; perpendicular to $\vec{T}$, points toward center of curvature.

Binormal: $\vec{B} = \vec{T} \times \vec{N}$; perpendicular to osculating plane; all three $\left(\vec{T}, \vec{N}, \vec{B}\right)$ are unit vectors forming a right-handed frame.

Torsion (rate of twisting out of the osculating plane; positive = twists toward $\vec{B}$):

$$\tau = -\frac{\mathrm{d}\vec{B}}{\mathrm{d}s} \cdot \vec{N} = \frac{(\vec{r}' \times \vec{r}'') \cdot \vec{r}'''}{|\vec{r}' \times \vec{r}''|^2}$$

5. Extrema

Second derivative test ($f : \mathbb{R}^2 \to \mathbb{R}$): at a critical point, let $D = f_{xx}f_{yy} - f_{xy}^2 = \det \mathrm{H}f$:

$D > 0$, $f_{xx} > 0$	$D > 0$, $f_{xx} < 0$	$D < 0$	$D = 0$
local min	local max	saddle	inconclusive

Lagrange multipliers: at a constrained extremum of f on $\{\vec{x} : g(\vec{x}) = k\}$, the level sets of f and g are tangent, so their gradients are parallel. Solve:

$$\nabla f(\vec{x}) = \lambda \nabla g(\vec{x}), \quad g(\vec{x}) = k$$

6. Vector calculus operators

Divergence: $\operatorname{div} \vec{F} = \nabla \cdot \vec{F} = \sum_i \frac{\partial F_i}{\partial x_i}$. Measures source density at each point: positive = net outflow, negative = net inflow, zero = incompressible/solenoidal.

Curl (3D): $\operatorname{curl} \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{vmatrix}$. Measures local rotation; direction = axis (right-hand rule), magnitude = angular speed.

Lemma: $\operatorname{div}\left(\operatorname{curl} \vec{F}\right) = 0$ for all $\vec{F}$.

2D curl: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ (the z-component of $\nabla \times \vec{F}$).

Conservative field test: on a simply connected domain, the following are equivalent:
- $\vec{F}$ is conservative
- $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all $i \neq j$
- $\operatorname{curl} \vec{F} = 0$ in 3D (irrotational)

The following are always equivalent:
- $\vec{F}$ is conservative
- $\vec{F} = \nabla f$ for some potential f
- $\oint_C \vec{F} \cdot \mathrm{d}\vec{r} = 0$ for all closed C.

Finding a potential function: integrate F_1 w.r.t. x_1 to get $f + g(x_2, ..., x_n)$; differentiate w.r.t. remaining variables and match against $F_2, F_3, ...$ to determine g.

7. Line integrals

Scalar line integral (integrate f over arc length):

$$\int_C f \, \mathrm{d}s = \int_a^b f \left| \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} \right| \mathrm{d}t$$

Vector line integral (work done by $\vec{F}$ along C):

$$\int_C \vec{F} \cdot \mathrm{d}\vec{r} = \int_a^b \vec{F} \cdot \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} \, \mathrm{d}t$$

FTLI: if $\vec{F} = \nabla f$, then $\int_C \vec{F} \cdot \mathrm{d}\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$ — work depends only on endpoints, not the path.

8. Multiple integrals and coordinate systems

Polar (2D): $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$

$\iint_D f \, \mathrm{d}A = \iint f \, r \, \mathrm{d}r \, \mathrm{d}\theta$

Cylindrical: same as polar + $z = z$; volume element $r \, \mathrm{d}z \, \mathrm{d}r \, \mathrm{d}\theta$

$\iiint f \, \mathrm{d}V = \iiint f \, r \, \mathrm{d}z \, \mathrm{d}r \, \mathrm{d}\theta$

Spherical (φ = angle from z-axis, $0 \le \varphi \le \pi$; θ = azimuthal):

$$x = \rho \sin\varphi \cos\theta \quad y = \rho \sin\varphi \sin\theta \quad z = \rho\cos\varphi$$
$$\rho^2 = x^2 + y^2 + z^2$$

$\iiint f \, \mathrm{d}V = \iiint f \, \rho^2 \sin\varphi \, \mathrm{d}\rho \, \mathrm{d}\theta \, \mathrm{d}\varphi$

Change of variables (transformation $T_{UX} : U \to X$, i.e. $\vec{x} = T_{UX}(\vec{u})$):

$$\int \cdots \int_X f \, \mathrm{d}x_1 \ldots \mathrm{d}x_n$$
$$= \int \cdots \int_U f \, |\det \mathrm{J}T_{UX}| \, \mathrm{d}u_1 \ldots \mathrm{d}u_n$$

$|\det \mathrm{J}T_{UX}|$ is the local area/volume stretching factor. For polar, cylindrical, spherical the standard Jacobian determinants give the r and $\rho^2 \sin\varphi$ factors above.

9. Surfaces

Normal to parametric surface $\vec{X}(u, v)$: $\vec{N} = \vec{X}_u \times \vec{X}_v$ (orientation from right-hand rule)

For graph $z = g(x, y)$: $\vec{X}_x \times \vec{X}_y = \langle -g_x, -g_y, 1 \rangle$, so $\left|\vec{X}_x \times \vec{X}_y\right| = \sqrt{g_x^2 + g_y^2 + 1}$

Scalar surface integral (integrate f over surface area):

$$\iint_S f \, \mathrm{d}S = \iint_D f \left|\vec{X}_u \times \vec{X}_v\right| \mathrm{d}A$$

Flux (net flow of $\vec{F}$ through oriented surface — sign depends on normal orientation):

$$\iint_S \vec{F} \cdot \mathrm{d}\vec{S} = \iint_D \vec{F} \cdot \left(\vec{X}_u \times \vec{X}_v\right) \mathrm{d}A$$

10. Parameterizations of common surfaces

Graph of $z = f(x, y)$:

$$\vec{X}(x, y) = \begin{bmatrix} x \\ y \\ f(x, y) \end{bmatrix} \quad \vec{X}_x \times \vec{X}_y = \begin{bmatrix} -f_x \\ -f_y \\ 1 \end{bmatrix}$$

Cylinder of radius R:

$$\vec{X}(\theta, z) = \begin{bmatrix} R\cos\theta \\ R\sin\theta \\ z \end{bmatrix}$$

Sphere of radius R:

$$\vec{X}(\varphi, \theta) = \begin{bmatrix} R\sin\varphi\cos\theta \\ R\sin\varphi\sin\theta \\ R\cos\varphi \end{bmatrix} \quad \left|\vec{X}_\varphi \times \vec{X}_\theta\right| = R^2 \sin\varphi$$

For an **ellipsoid** with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}$, scale the outputs of sphere $\vec{X}$ by a, b, c.

Cone $z = cr$:

$$\vec{X}(r, \theta) = \begin{bmatrix} r\cos\theta \\ r\sin\theta \\ cr \end{bmatrix}$$

Surface of revolution of $f(x)$ around the x-axis:

$$\vec{X}(u, v) = \begin{bmatrix} u \\ f(u)\cos v \\ f(u)\sin v \end{bmatrix}$$

Variable bounded by function: If z (in the output coordinate space) is bounded by $z \in [0, h(u)]$, define $z := vh(u)$ with $v \in [0, 1]$ to make the uv-domain a rectangle.

11. The big three theorems

Green's, Stokes', and the Divergence Theorem each relate an integral over a region to an integral over its boundary, reducing dimension by one.

Green's ($C = \partial D$, counterclockwise; D a planar region): circulation around the boundary = total 2D curl (rotation density) in the interior:

$$\oint_{\partial D} P \, \mathrm{d}x + Q \, \mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}A$$

Stokes' ($C = \partial S$, consistently oriented; S a surface): circulation around the boundary = total 3D curl through any spanning surface:

$$\oint_{\partial S} \vec{F} \cdot \mathrm{d}\vec{r} = \iint_S \left(\nabla \times \vec{F}\right) \cdot \mathrm{d}\vec{S}$$

S can be replaced by any other surface sharing boundary C — pick whichever is easier to integrate over.

Green's is equivalent to Stokes' with the surface being in the xy-plane and the normal vector in the $+z$-direction.

For Green's, regions with holes can be done by orienting the outer boundary counterclockwise and the inner boundary clockwise; for Stokes' make sure they are oriented according to right-hand rule with thumb as the normal vector.

Divergence ($S = \partial E$, outward normals; E a solid region): net outward flux through a closed surface = total divergence (source density) in the enclosed region:

$$\oiint_{\partial E} \vec{F} \cdot \mathrm{d}\vec{S} = \iiint_E \nabla \cdot \vec{F} \,\mathrm{d}V$$

Sign check: positive net outward flux $\leftrightarrow$ net positive sources inside.

12. Electromagnetism

$\vec{E}$-field: It points outwards from positive charge, inward towards negative charge, with magnitude

$$\left|\vec{E}\right| = \frac{kq}{r^2} = \frac{q}{4\pi\varepsilon_0}\frac{1}{r^2}.$$

Gauss's law: With outward-pointing (positively-oriented) $d\vec{S}$:

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \quad \oiint_S \vec{E} \cdot d\vec{S} = \frac{q_{\text{enc}}}{\varepsilon_0}$$

13. Random trig

°	rad	sin	cos	tan
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	undef

$$\cos^2 x + \sin^2 x = 1.$$